

Denoiser	Mean test accuracy (%)	Test range (%)	Bandpass deficit (pp)	Bandpass lower in matched cells
wavelet	83.78	83.03–85.00	43.68	16/16
butterworth	83.46	81.97–85.00	43.36	16/16
savgol	83.71	81.97–85.15	43.62	16/16
hampel	82.72	81.06–84.09	42.62	16/16
bandpass	40.09	36.67–43.18	--	--

Five denoisers × 16 matched downstream configurations (80 runs total).
All 16 bandpass configurations ranked last within their matched cells; all runs used seed 42.